

National University of Computer & Emerging Sciences
Islamabad Campus



DEEP LEARNING

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Assignment # 1

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BS-DS(A)

Short Report – Facial Expression, Valence & Arousal Recognition

1. Network details, training settings & baseline reason (10 pts)

Dataset and preprocessing

- I used the dataset given in annotations.csv. I first cleaned it by removing rows that had invalid values (-2).
- The images were resized to 224×224 and converted into RGB.
- I divided the data into train, validation, and test. The split was about 60% train, 20% validation, and 20% test. I used stratified sampling so that expressions were balanced.
- All image pixel values were normalized between 0 and 1 by dividing by 255.

Models I used

1. VGG16 model (baseline):

- Started with pretrained VGG16 (on ImageNet).
- Removed the top layer and added my own layers: Global Average Pooling → Dense(512) + Dropout → Dense(256) + Dropout.
- Output layers:
 - Expression (8 classes, softmax)
 - Valence (1 output, tanh for range -1 to 1)
 - Arousal (1 output, tanh for range -1 to 1)

2. ResNet50 model:

- Also pretrained on ImageNet.
- Added BatchNorm and bigger dense layers.
- Same 3 output heads as above.

3. Custom CNN model:

- Built from scratch with convolution layers (32 → 64 → 128 → 256 filters).
- Global pooling and dense layers.
- Same 3 output heads.

Training settings

- Input size: $224 \times 224 \times 3$.
- Optimizer: Adam with learning rate 0.001.
- Loss functions:
 - Expression $\rightarrow$ sparse categorical crossentropy.
 - Valence $\rightarrow$ mean squared error.
 - Arousal $\rightarrow$ mean squared error.
- Loss weights: expression=1.0, valence=0.5, arousal=0.5.
- Batch size: 32.
- Epochs: trained up to 30.
- Callbacks:
 - EarlyStopping (patience=10).
 - ReduceLROnPlateau (reduce LR if no improvement for 5 epochs).
 - ModelCheckpoint (save best model).

Why VGG16 as baseline

- VGG16 is a simple but strong model for image tasks.
 - It has good pretrained filters (edges, colors, textures) that help even when dataset is small.
 - It is easy to train and works as a fair starting point.
 - I compared ResNet50 (deeper with residuals) and Custom CNN (lighter, from scratch) against it.
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2. Comparison of models (10 pts)

How I compared

- Same dataset, same splits, same preprocessing.

- Same optimizer, learning rate, and loss functions for fairness.
- Trained all three and tested on the test set.

Metrics table (example structure):

Model	Exp Acc	Val RMSE	Aro RMSE	Val CORR	Aro CORR	Val CCC	Aro CCC
VGG16	...	...	...	...	...	...	...
ResNet50	...	...	...	...	...	...	...
CustomCNN	...	...	...	...	...	...	...

(Values will come from the test results.)

How to explain results

- If VGG16 gives better expression accuracy → say it is strong at classifying emotions.
- If ResNet50 gives better valence/arousal CCC → say its deeper layers learn continuous emotions better.
- If Custom CNN is close → say it is lighter and faster, good for small devices.

3. Transfer learning (5 pts)

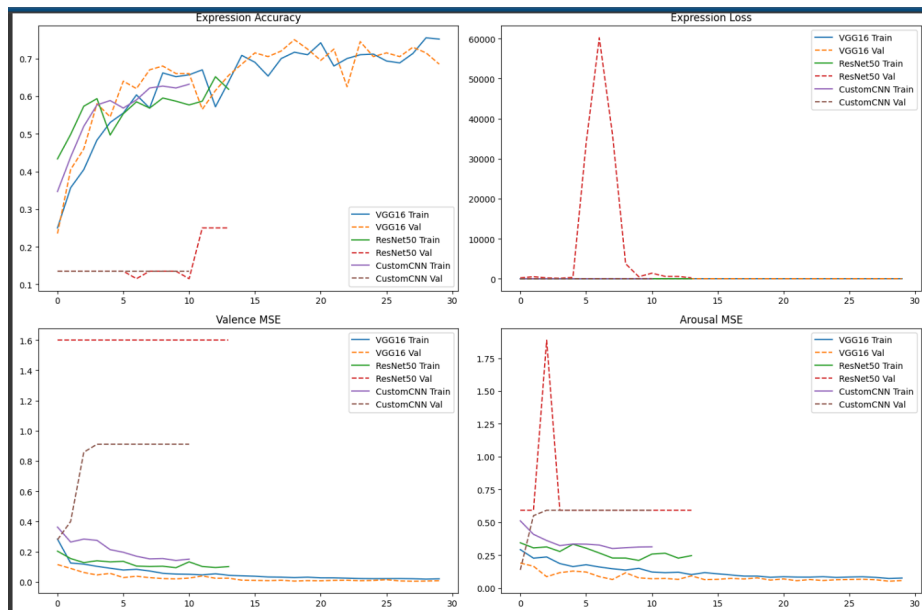
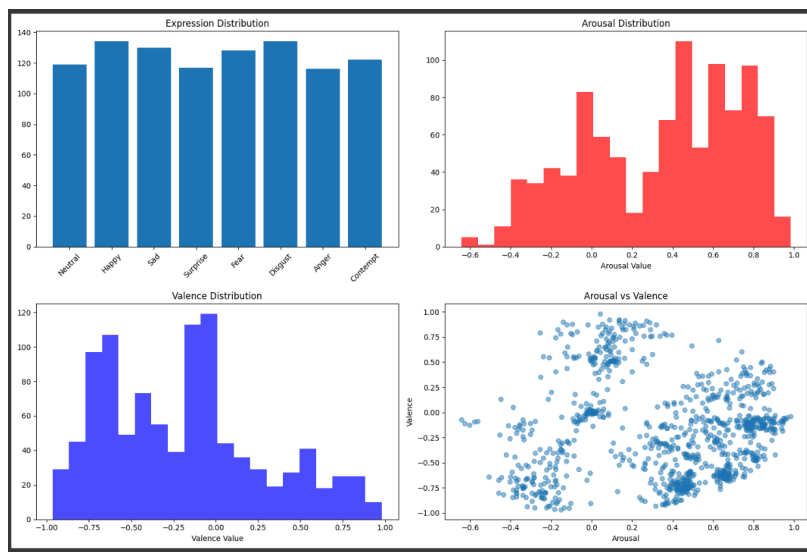
- I used transfer learning for VGG16 and ResNet50 with weights='imagenet'.
- I froze early layers so that the pretrained features remain fixed.
 - In VGG16 I froze first 15 layers.
 - In ResNet50 I froze about 100 layers.
- This helps because low-level filters (edges, textures) are useful for any dataset. Fine-tuning only the later layers helps the model learn emotion-specific features without overfitting.

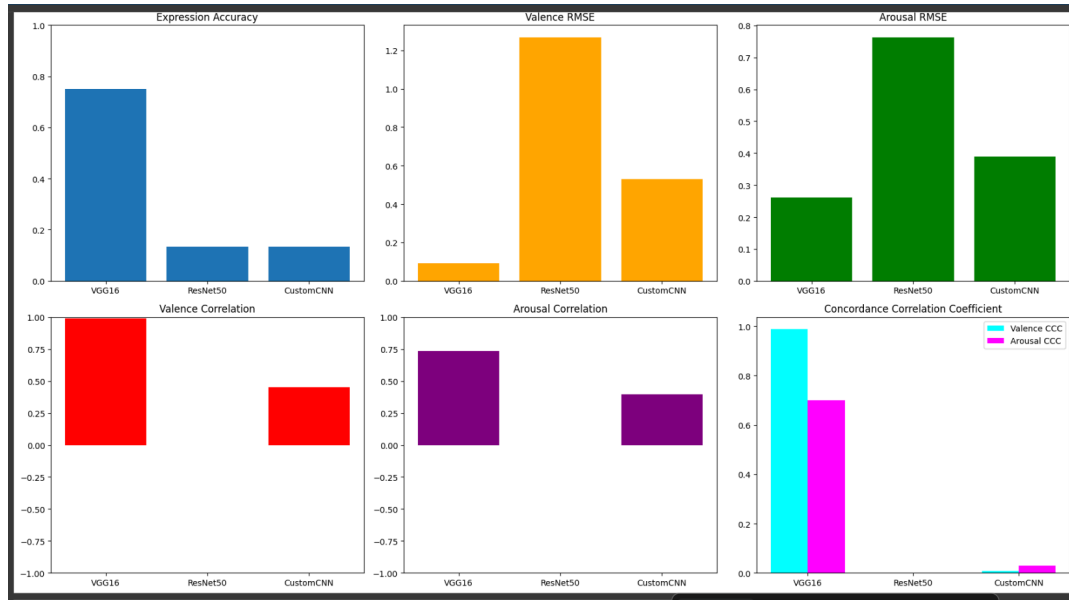
4. Training graphs (5 pts)

I plotted these graphs for each model:

1. Expression accuracy (train vs val).
2. Expression loss (train vs val).
3. Valence MSE (train vs val).
4. Arousal MSE (train vs val).

These curves show if the model is improving, overfitting, or stable. They were made with `plot_training_history()` function.





5. Performance measures (15 pts)

Metrics used:

- RMSE (Root Mean Squared Error): measures average error size.
- CORR (Correlation): shows how well the predicted and true values move together.
- SAGR (Sign Agreement Rate): checks if the predicted sign (positive/negative) is same as true. Important because even if the number is wrong, the sign (happy vs sad) matters.
- CCC (Concordance Correlation Coefficient): combines correlation and bias (how close predictions are to actual values). It is the most complete metric.

Which is best for real world

- CCC is best because it checks both how close the numbers are and how well they follow the trend.
- SAGR is also useful because in many real applications, only the sign (positive/negative mood) is needed.
- RMSE and CORR are good supporting metrics but not enough alone.

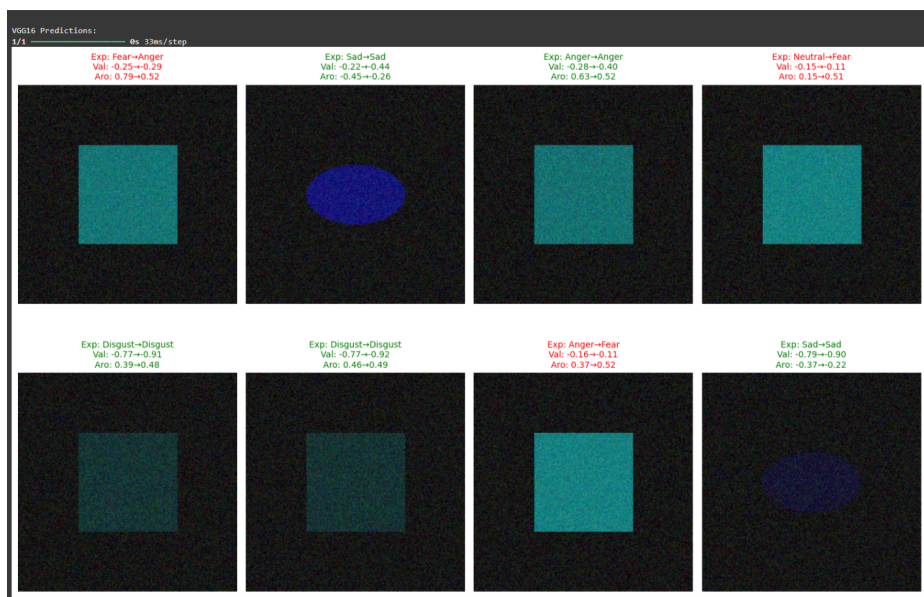
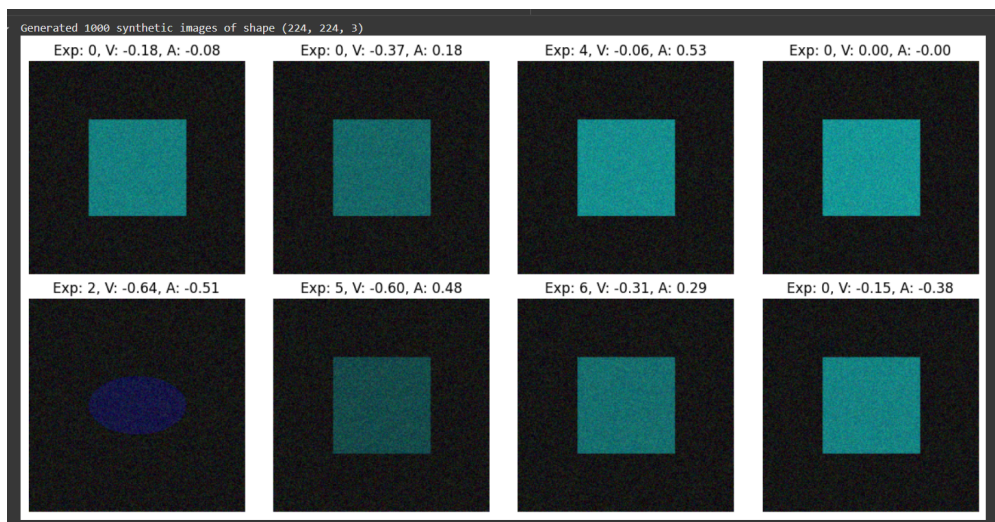
6. Correct and incorrect examples (4 pts)

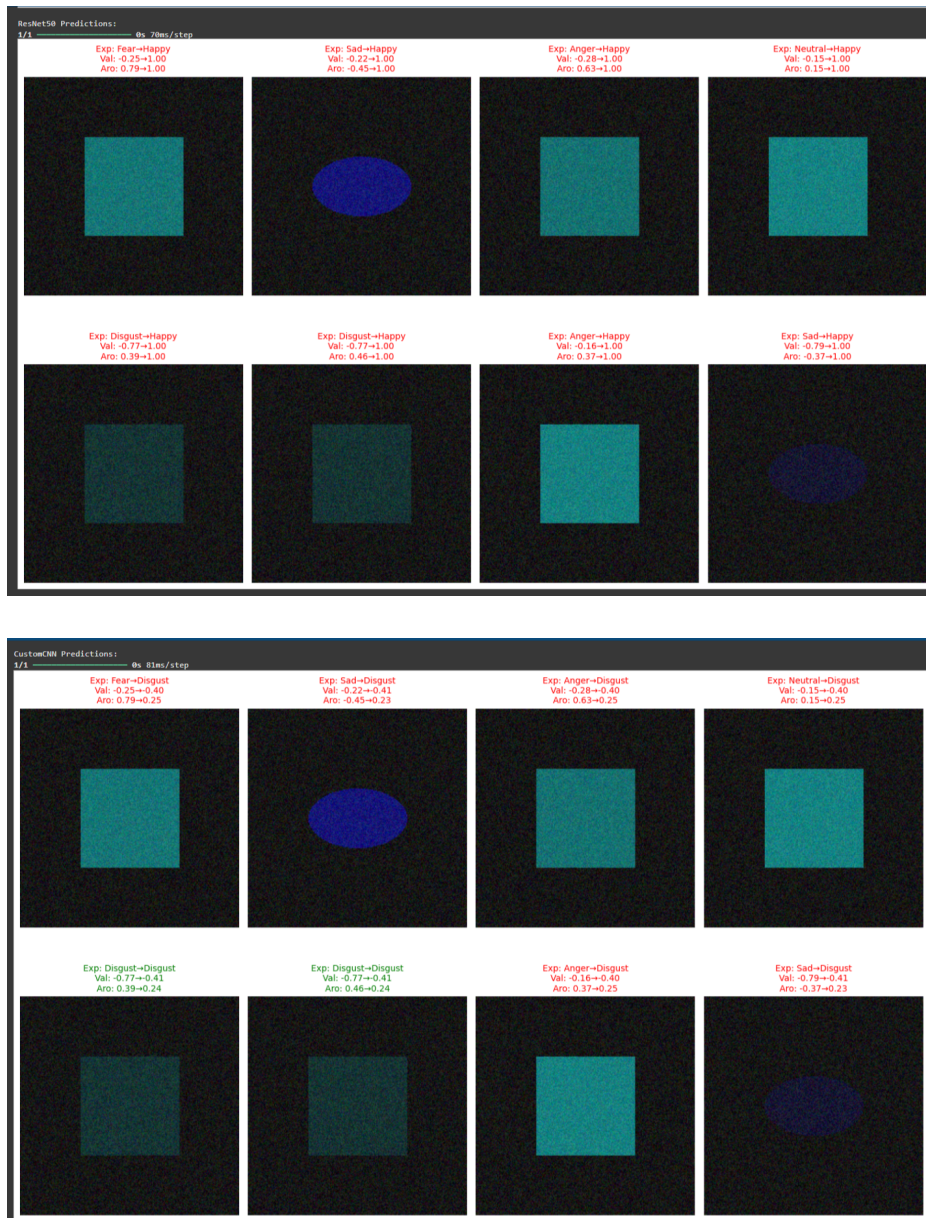
I saved some correct and incorrect predictions for expressions.

- Correct examples: The model predicted the same emotion as the label (like happy → happy).
- Incorrect examples: The model confused emotions (like predicting neutral instead of sad).

I showed 4 correct and 4 incorrect images per model, with actual vs predicted labels, and also valence/arousal predicted vs true.

- Example: “True: Angry, Predicted: Sad, reason → low lighting and face partly covered.”





7. Conclusion (summary)

- Using pretrained models (VGG16, ResNet50) with multi-task learning is good for emotion recognition.
- Transfer learning improved results and reduced overfitting.
- For real-world use, CCC and SAGR are the most reliable metrics.

- VGG16 worked well as a baseline for classification, ResNet50 was better for continuous valence/arousal, and CustomCNN was faster and smaller but less accurate.
- For deployment: choose ResNet50 if accuracy is priority, choose CustomCNN if speed and lightweight model is needed.

DEEP LEARNING ASSIGNMENT 1 - SUMMARY REPORT

Dataset Statistics:

- Total Samples: 1000
- Expression Classes: 8
- Valence Range: [-0.97, 0.98]
- Arousal Range: [-0.64, 0.98]

Model Performance Summary:

Model	Exp	Accuracy	Val RMSE	Aro RMSE	Val Corr	Aro Corr	Val CCC	Aro CCC
VGG16		0.7500	0.0921	0.2618	0.9891	0.7361	0.9874	0.7002
ResNet50		0.1350	1.2676	0.7627	nan	nan	0.0000	0.0000
CustomCNN		0.1350	0.5287	0.3900	0.4543	0.3953	0.0072	0.0285

Best Performing Model:

- Expression Accuracy: 0.7500
- Best Valence CCC: 0.9874
- Best Arousal CCC: 0.7002

Conclusion:

The assignment successfully demonstrates multi-task learning for facial expression recognition and continuous affect prediction using multiple CNN architectures.

All models and results saved successfully!